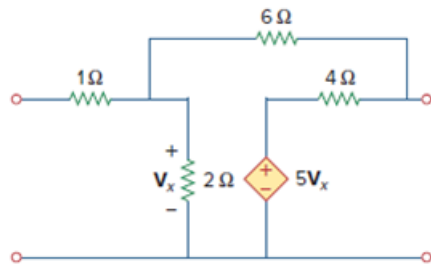


Problem

Rework Prob. 19.47 using *PSpice* or *MultiSim*.

Obtain the **ABCD** parameters for the network in Fig. 19.102.

Figure 19.102.



Step-by-step solution

Step 1 of 15

Consider the following circuit diagram:

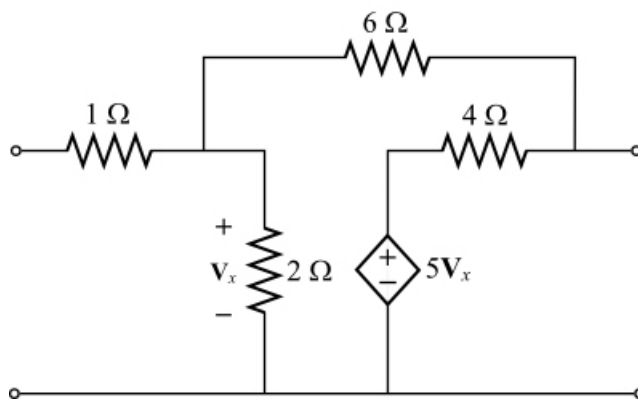


Figure 1

Step 2 of 15

Write the ABCD parameters.

$$\mathbf{A} = \left. \frac{\mathbf{V}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0} \quad \dots (1)$$

$$\mathbf{B} = - \left. \frac{\mathbf{V}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \quad \dots (2)$$

$$\mathbf{C} = \left. \frac{\mathbf{I}_1}{\mathbf{V}_2} \right|_{\mathbf{I}_2=0} \quad \dots (3)$$

$$\mathbf{D} = - \left. \frac{\mathbf{I}_1}{\mathbf{I}_2} \right|_{\mathbf{V}_2=0} \quad \dots (4)$$

Step 3 of 15

To calculate the values of **A** and **C** parameters, connect a test voltage source to input port and open circuit the output port. Draw the modified circuit diagram.

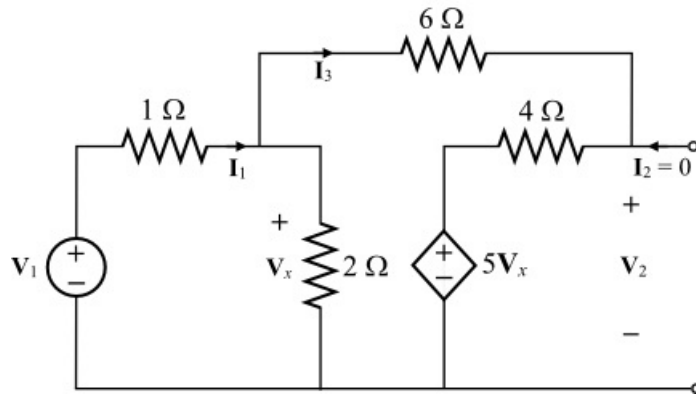


Figure 2

Step 4 of 15

From the Figure 2,

$$V_x = 2(I_1 - I_3)$$

Apply Kirchhoff's voltage law to the left side loop.

$$\begin{aligned} V_1 &= I_1 + 2(I_1 - I_3) \\ &= 3I_1 - 2I_3 \dots\dots (5) \end{aligned}$$

Apply Kirchhoff's current law to middle side loop.

$$\begin{aligned} -V_x + (6 + 4)I_3 + 5V_x &= 0 \\ 10I_3 + 4[2(I_1 - I_3)] &= 0 \quad \left(\text{since } V_x = 2(I_1 - I_3) \right) \\ 8I_1 + 2I_3 &= 0 \\ I_3 &= -4I_1 \dots\dots (6) \end{aligned}$$

Step 5 of 15

Apply Kirchhoff's current law to right side loop.

$$\begin{aligned} V_2 &= 5V_x + 4I_3 \\ &= 5[2(I_1 - I_3)] + 4I_3 \\ &= 10I_1 - 6I_3 \end{aligned}$$

Substitute $-4I_1$ for I_3 in the above equation.

$$\begin{aligned} V_2 &= 10I_1 - 6(-4I_1) \\ &= 10I_1 + 24I_1 \\ &= 34I_1 \end{aligned}$$

Substitute $-4I_1$ for I_3 in the equation (5).

$$\begin{aligned} V_1 &= 3I_1 - 2[-4I_1] \\ &= 3I_1 + 8I_1 \\ &= 11I_1 \end{aligned}$$

Substitute $11I_1$ for V_1 and $34I_1$ for V_2 in the equation (1).

$$\begin{aligned} A &= \frac{V_1}{V_2} \\ &= \frac{11I_1}{34I_1} \\ &= 0.3235 \end{aligned}$$

Substitute $34I_1$ for V_2 in the equation (3).

$$\begin{aligned} C &= \frac{I_1}{V_2} \\ &= \frac{I_1}{34I_1} \\ &= 0.02941 \text{ S} \end{aligned}$$

Step 6 of 15

To calculate the values of **B** and **D** parameters, connect a test current source to input port and short circuit the output port. Draw the modified circuit diagram.

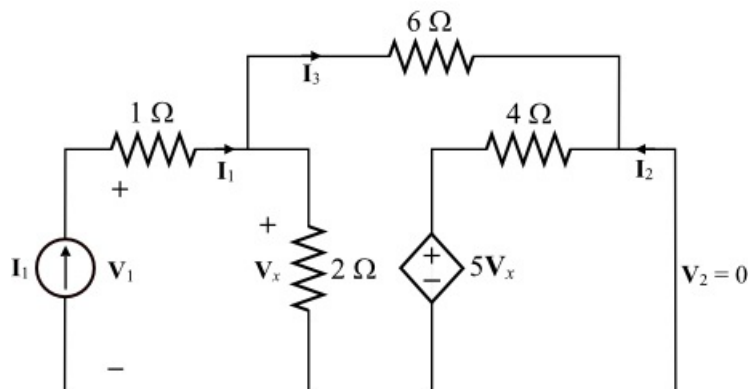


Figure 3

Step 7 of 15

From the Figure 3,

$$V_x = 2(I_1 - I_3)$$

Apply Kirchhoff's voltage law to the left side loop.

$$V_1 = I_1 + 2(I_1 - I_3)$$

$$= 3I_1 - 2I_3 \dots\dots (7)$$

Apply Kirchhoff's current law to middle side loop.

$$2(I_3 - I_1) + (6)I_3 + 4(I_3 + I_2) + 5V_x = 0$$

$$-2I_1 + 4I_2 + (2 + 6 + 4)I_3 + 5V_x = 0$$

$$-2I_1 + 4I_2 + (2 + 6 + 4)I_3 + 5[2(I_1 - I_3)] = 0 \quad (\text{since } V_x = 2(I_1 - I_3))$$

$$(-2 + 10)I_1 + 4I_2 + (2 + 6 + 4 - 10)I_3 = 0$$

$$8I_1 + 4I_2 + 2I_3 = 0 \dots\dots (8)$$

Step 8 of 15

Apply Kirchhoff's current law to right side loop.

$$5V_x + 4(I_2 + I_3) = 0$$

$$5[2(I_1 - I_3)] + 4I_2 + 4I_3 = 0$$

$$10I_1 + 4I_2 + (-10 + 4)I_3 = 0$$

$$10I_1 + 4I_2 - 6I_3 = 0$$

$$I_3 = \frac{5I_1 + 2I_2}{3} \dots\dots (9)$$

Substitute $\frac{5I_1 + 2I_2}{3}$ for I_3 in the equation (8).

$$8I_1 + 4I_2 + 2\left(\frac{5I_1 + 2I_2}{3}\right) = 0$$

$$24I_1 + 12I_2 + 10I_1 + 4I_2 = 0$$

$$34I_1 + 16I_2 = 0$$

$$I_2 = -\frac{34I_1}{16}$$

Step 9 of 15

Substitute $\frac{5\mathbf{I}_1 + 2\mathbf{I}_2}{3}$ for $\mathbf{I}_3$ in the equation (7).

$$\begin{aligned}\mathbf{V}_1 &= 3\mathbf{I}_1 - 2\left[\frac{5\mathbf{I}_1 + 2\mathbf{I}_2}{3}\right] \\ &= \frac{9\mathbf{I}_1 - 10\mathbf{I}_1 - 4\mathbf{I}_2}{3} \\ &= \frac{-\mathbf{I}_1 - 4\mathbf{I}_2}{3}\end{aligned}$$

Now substitute $\frac{34\mathbf{I}_1}{16}$ for $\mathbf{I}_2$ in the above equation.

$$\begin{aligned}\mathbf{V}_1 &= \frac{-\mathbf{I}_1 - 4\left(\frac{34\mathbf{I}_1}{16}\right)}{3} \\ &= \frac{16\mathbf{I}_1 - 136\mathbf{I}_1}{48} \\ &= \frac{-5\mathbf{I}_1}{2}\end{aligned}$$

Substitute $\frac{-5\mathbf{I}_1}{2}$ for $\mathbf{V}_1$ and $\frac{34\mathbf{I}_1}{16}$ for $\mathbf{I}_2$ in the equation (2).

$$\begin{aligned}\mathbf{B} &= -\frac{\mathbf{V}_1}{\mathbf{I}_2} \\ &= -\frac{\frac{-5\mathbf{I}_1}{2}}{\frac{34\mathbf{I}_1}{16}} \\ &= \frac{(5)(16)}{68} \\ &= 1.176\Omega\end{aligned}$$

Substitute $\frac{34\mathbf{I}_1}{16}$ for $\mathbf{I}_2$ in the equation (4).

$$\begin{aligned}\mathbf{D} &= -\frac{\mathbf{I}_1}{\mathbf{I}_2} \\ &= -\frac{\mathbf{I}_1}{\frac{34\mathbf{I}_1}{16}} \\ &= \frac{16}{34} \\ &= 0.4706\end{aligned}$$

Therefore, the transmission parameters, $\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}$ are $\boxed{\begin{bmatrix} 0.3235 & 1.176\Omega \\ 0.12941\text{S} & 0.4706 \end{bmatrix}}$.

PSpice analysis:

- Open the CIS Demo software.
- Once you are in the capture session frame, click on the menu item File, select new, and then click on Project.
- In the New Project box, type 'selected file name' in the Name text box. Ensure that the Analog or Mixed-signal circuit Wizard is activated.
- Connected the circuit diagram as per the Circuit.
- Double click on each component and enter the given values.
- Click the New Simulation Profile icon and enter a name in the Name text box. You will need to enter the appropriate settings for this project in the Simulation
- Setting box. Click the Analysis tab, and select Bias Point from the Analysis type list. Click OK and save the document.
- Click on the run icon. You will observe the A/D Demo screen. When you close this screen, you will observe the bias voltages and currents.

Step 11 of 15

To calculate the values of **A** and **C** parameters, connect a 1 V test voltage source to input port and open circuit the output port.

The following is the PSpice schematic:

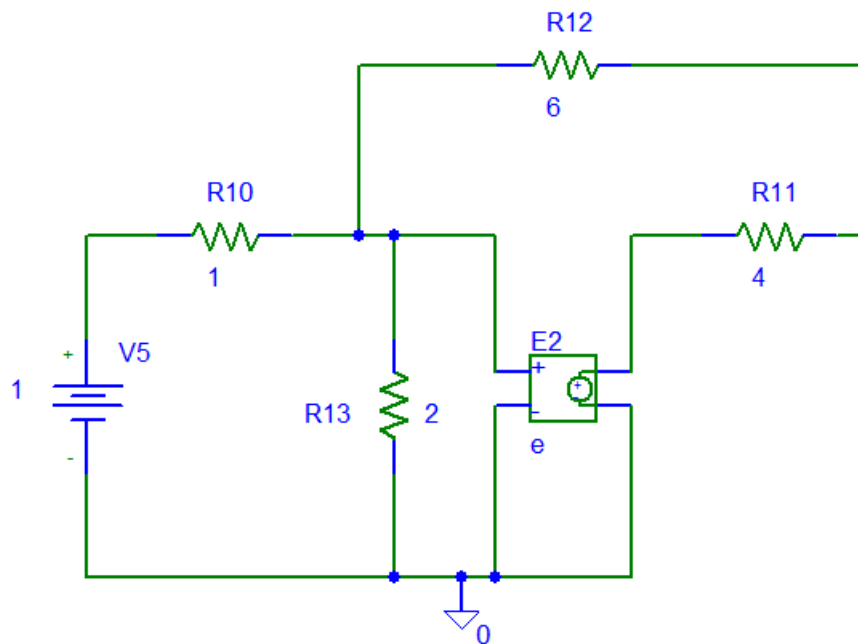


Figure 4

Step 12 of 15

Run the Schematic, the following is the schematic circuit with voltage and current values.

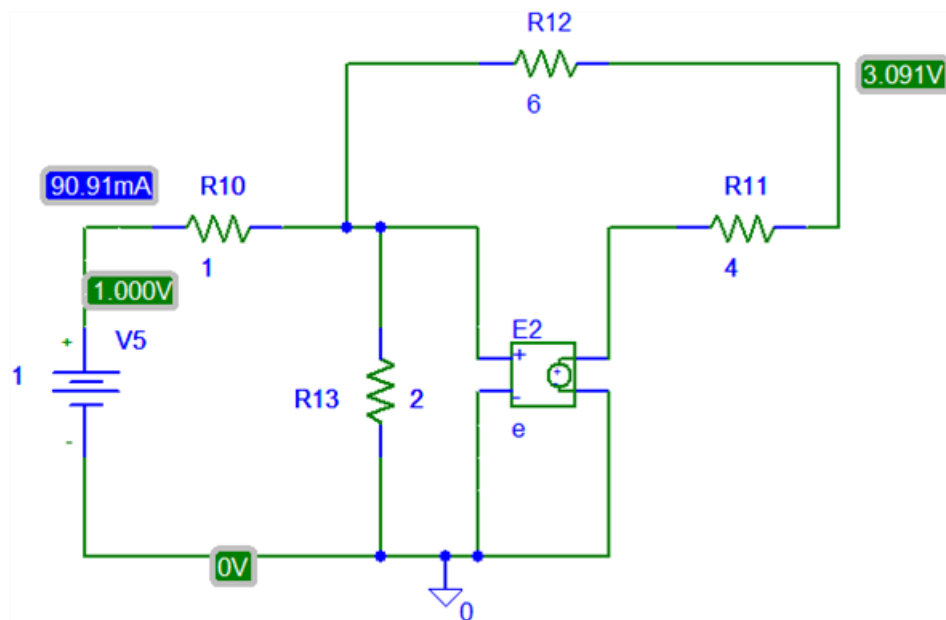


Figure 5

From the Figure 5,

$$I_1 = 90.91 \text{ mA}$$

$$I_2 = 0 \text{ A}$$

$$V_1 = 1 \text{ V}$$

$$V_2 = 3.091 \text{ V}$$

Substitute 1 V for V_1 and 3.091 V for V_2 in the equation (1).

$$\begin{aligned} A &= \frac{V_1}{V_2} \\ &= \frac{1}{3.091} \\ &= 0.3235 \end{aligned}$$

Substitute 90.91 mA for I_1 and 3.091 V for V_2 in the equation (3).

$$\begin{aligned} C &= \frac{I_1}{V_2} \\ &= \frac{90.91 \times 10^{-3}}{3.091} \\ &= 0.02941 \text{ S} \end{aligned}$$

To calculate the values of **B** and **D** parameters, connect a 1 A test current source to input port and short circuit the output port.

The following is the PSpice schematic:

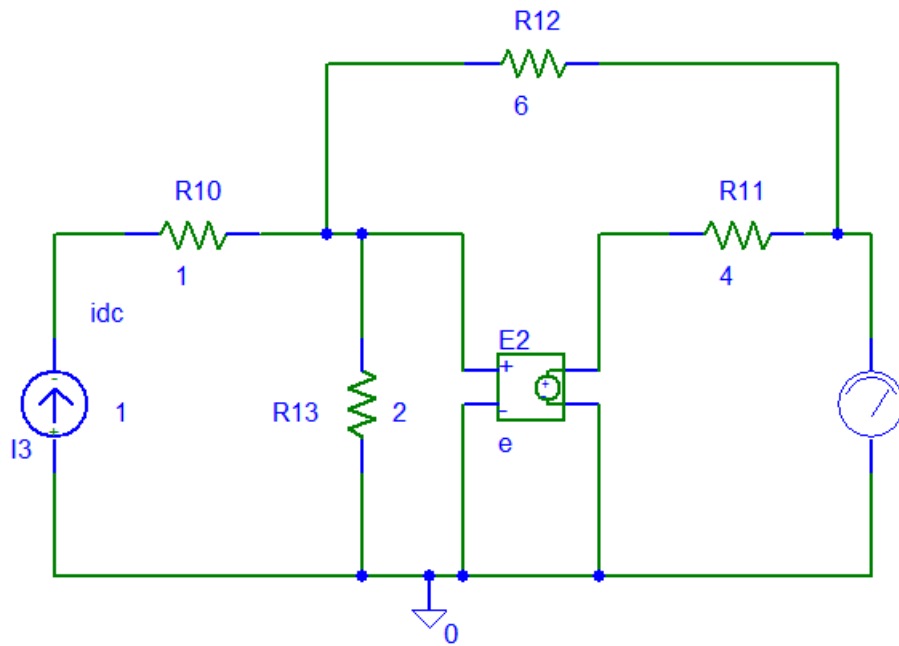


Figure 6

Step 14 of 15

Run the Schematic, the following is the schematic circuit with voltage and current values.

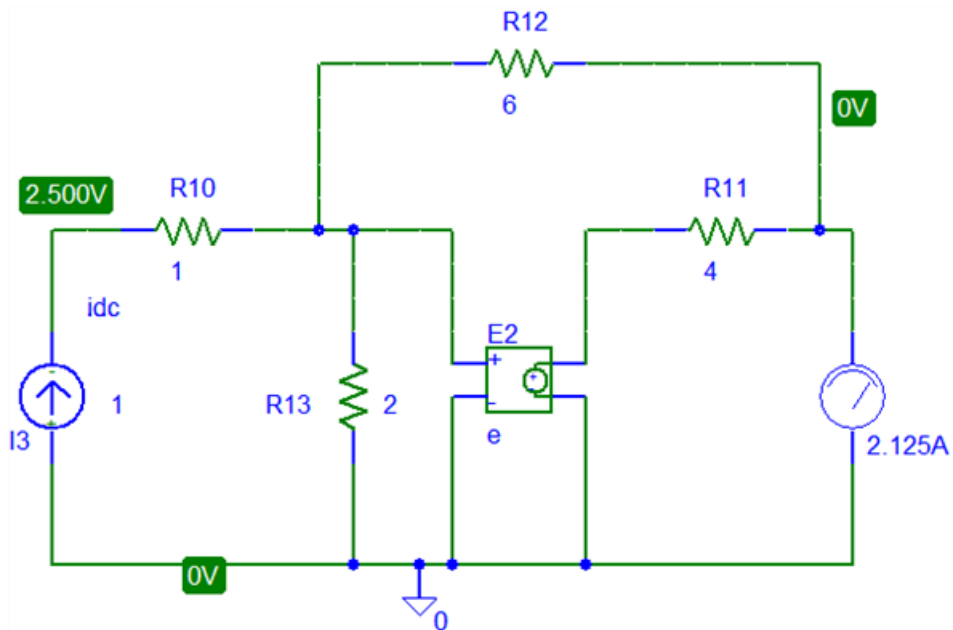


Figure 7

Step 15 of 15

From the Figure 7,

$$I_1 = 1 \text{ A}$$

$$I_2 = -2.215 \text{ A}$$

$$V_1 = 2.5 \text{ V}$$

$$V_2 = 0 \text{ V}$$

Substitute 2.5 V for V_1 and -2.215 A for I_2 in the equation (2).

$$\begin{aligned} B &= -\frac{V_1}{I_2} \\ &= -\frac{2.5}{-2.215} \\ &= 1.176 \Omega \end{aligned}$$

Substitute 1 A for I_1 and -555.56 mA for I_2 in the equation (4).

$$\begin{aligned} D &= -\frac{I_1}{I_2} \\ &= -\frac{1}{-2.125} \\ &= 0.4705 \end{aligned}$$

Therefore, the transmission parameters, $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$ are $\begin{bmatrix} 0.3235 & 1.176 \Omega \\ 0.02941 \text{ S} & 0.4705 \end{bmatrix}$.